MILITARY SPECIFICATION SHEET

CABLES, RADIO FREQUENCY, COAXIAL, 93 OHMS, M17/90-RG71

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the cables described herein shall consist of this document and the latest issue of Specification MIL-C-17.

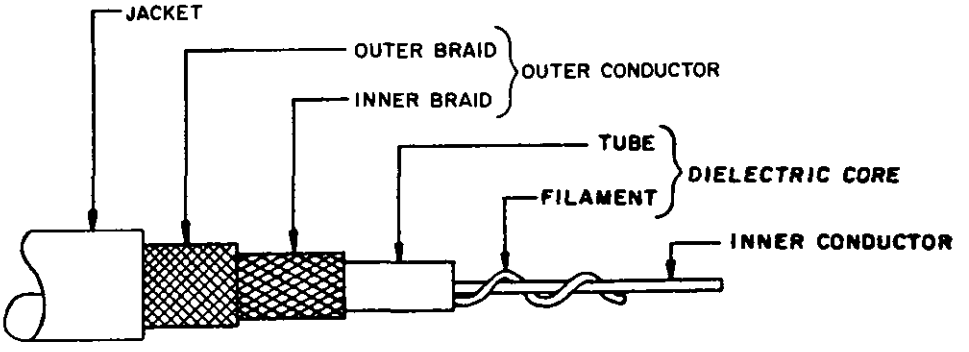


FIGURE 1. Configuration.

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TABLE I. Description.

Components	Construction details		
Inner conductors	Solid, copper-covered, steel wire. Diameter: .0253 inch ±.0010.		
Dielectric core	Type A-3: Air-spaced polyethylene. A monofilament thread, .035 inch approximate diameter with a lay of 1/2 inch approximate, under an extruded tube. Diameter: .146 inch ±.005.		
		Alternate	
	A continuous tube, .003 inch thick maximum, under two continuous spiral fins, with a lay of 1-1/4 inches approximate, under an extruded tube.		
Outer conductor	Double braid of copper wire. Diameter: .208 inch, maximum.		
		Alternate	
Inner braid	Wire gauge:	AWG #34	AWG #36
	Wire coating:	Bare	Tinned
	Coverage:	94.3% nominal	95.2% nominal
	Carriers:	16	24
	Ends:	7	6
	Picks/inch:	8.2 ±10%	12.0 ±10%
Outer braid	Wire gauge:	AWG #36	AWG #36
	Wire coating:	Tinned	Tinned
	Coverage:	93.8% nominal	94.6% nominal
	Carriers:	16	24
	Ends:	9	6
	Picks/inch:	10.6 ±10%	16.0 ±10%
Jacket	Type IIIa Diameter: .245 inch ±.005.		

Engineering information:

Continuous working voltage: 750 Vrms, maximum.
Operating frequency: 1 GHz, maximum.
Velocity of propagation: 84 percent, nominal.
Power rating: See figure 2.
Operating temperature range: -55° to +85°C.
Inner conductor properties:
DC resistance (maximum at 20°C): 4.4 ohms per 100 feet.
Elongation: 1 percent, minimum.
Tensile strength: 110 $klb_f/inch^2$, minimum.
Engineering notes: This cable useful in low temperature applications (see connector series TNC or BNC per MIL-C-39012).

REQUIREMENTS:

Dimensions, configuration, and descriptions: See figure 1 and table I.

Environmental and mechanical:
Adhesion of conductors: Not applicable.
Aging stability: +90° ±2°C.
Stress crack resistance: Not applicable.
Outer conductor integrity: Not applicable.
Cold bend: -55 ±2°C.
Dimensional stability: +85° ±2°C.
Inner conductor from core: .062 inch, maximum.
Inner conductor from jacket: .125 inch, maximum.
Contamination: Not applicable.
Bendability: Not applicable.
Flammability: Not Applicable.
Weight: 5 pounds per 100 feet, maximum.

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Electrical:
Test frequency: 50 MHz to 1 GHz.
Spark test: 5,000 Vrms, +25%, -0%.
Voltage withstanding: 3,000 Vrms, minimum.
Insulation resistance: Not applicable.
Corona extinction voltage: Not applicable.
Characteristic impedance: 93 ohms ±5.
Attenuation: 8.0 dB per 100 feet at 400 MHz.
Structural return loss: Not applicable.
Capacitance: 14.5 pF per foot, maximum.
Capacitance stability: ±1.5 percent.
Capacitance unbalance: Not applicable.
Transmission unbalance: Not applicable.
Mechanically induced noise voltage: Not applicable.
Time delay: Not applicable.

Part number: See table II.

Supersession data: See table II.

TABLE II. Cross reference of part number.

Part number	Superseded part number or type designation
M17/90-RG71	RG71B/U

NOTE: Revision letters are not used to denote changes due to the extensiveness of the changes.

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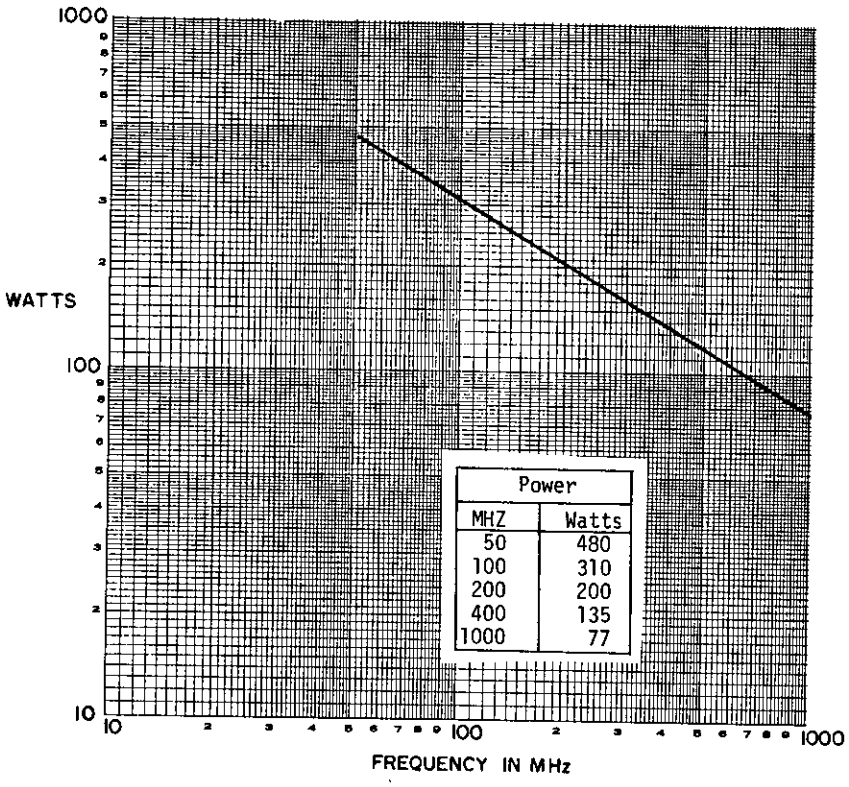


FIGURE 2. Power rating

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Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - SH
Air Force - 11, 17, 99
DLA - ES, IS

User activities:
Army - ME, AT, AR
Navy - AS, OS, MC
Air Force - 19

Preparing activity:
Army - CR

Agent:
DLA - ES

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